

On the relationship between induced polarization and surface area in metal-sand and clay-sand mixtures

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ABSTRACT

Induced polarization (IP) measurements were conducted on saturated kaolinite-, iron-, and magnetite-sand mixtures as a function of varying percentage weight of a mineral constituent: 0%–100% for iron and magnetite and 0%–32% for kaolinite. We determined the specific surface area for each mineral using nitrogen gas adsorption, where the porosity of each mixture was calculated from weight loss after drying. We fit a Cole-Cole model (Cole and Cole, 1941) to the electrical data obtained for the magnetite and iron mixtures. In contrast, the kaolinite mixtures showed a power-law dependence of phase on frequency. The global polarization magnitude we obtained from the Cole-Cole modeling of the iron and magnetite mixtures displays a single, near-linear dependence on the ratio of surface area to pore volume (S_p) calculated for the mixtures. A similar relationship is found using a local measure of polarization (imaginary conductivity at 1 Hz) for the clay-sand mixtures. The S_p appears to be a critical parameter for determining IP in both metallic- and clay-containing soils. This result is not easily reconciled with traditional models of induced polarization.

INTRODUCTION

Interest is growing in the application of induced polarization (IP) for the investigation of environmental processes. The IP signal results from a diffusive and/or electrochemical polarization occurring within the electrical double layer (EDL) at the mineral-fluid interface. The physicochemical, interfacial properties controlling IP are important parameters for understanding environmental processes such as groundwa-

ter flow, solute retardation, chemical alteration (precipitation/dissolution), and microbial-mineral interactions.

The need to better understand environmental processes has stimulated research on the physicochemical controls on IP in soils. Lesmes and Frye (2001) studied the relationship between IP measurements recorded with geophysical instrumentation and electrical parameters defining polarization. Their research shows that imaginary conductivity (the effective dielectric permittivity scaled by the measurement frequency) is directly related to the interfacial properties, whereas measured IP parameters (phase angle and chargeability) reflect both polarization and conduction processes occurring in soils.

The purpose of our study is to improve understanding of the dependence of IP on interfacial properties of soils. Studies show that the mineral surface area in contact with fluid determines the IP response of soils and rocks: Power law relationships between imaginary conductivity and specific surface area normalized to pore volume (S_p) (Börner and Schön, 1991) (Börner et al., 1996) (Slater and Glaser, 2003) are reported. However, this observation appears inconsistent with traditional models of induced polarization that describe the IP responses of metallic mineral deposits and clay soils.

We examine the dependence of IP on the interfacial surface in contact with pore fluid for (1) a metal-sand mixture exhibiting electrode polarization and (2) a clay-sand mixture wherein membrane polarization might occur. The relationship between IP magnitude and S_p for both systems is compared with relationships observed for natural soils in recent literature. Our results suggest that S_p is the critical structural parameter determining IP — irrespective of mineralogy, i.e., metal or clay.

METHODS

Synthetic samples were created by mixing Ottawa sand (0.6–0.7 mm in diameter, density (ρ) = 2.64 g/cm³ with (1) iron filings (mean grain size 0.4 mm; ρ = 3.29 g/cm³, mean aspect

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ratio of 5:1); (2) magnetite sand (mean grain size 0.43 mm; $\rho = 4.7 \text{ g/cm}^3$; trace amounts of quartz, ilmenite, zircon, garnet); and (3) kaolinite clay (mean grain size 0.002 mm, $\rho = 2.6 \text{ g/cm}^3$). We used a nitrogen gas-adsorption BET method (Brunauer et al., 1938) — repeated three times — to determine for a single sample the specific surface area (S_s) of the Ottawa sand ($0.0485 \pm 0.0015 \text{ m}^2/\text{g}$), iron ($2.84 \pm 0.09 \text{ m}^2/\text{g}$), magnetite ($1.08 \pm 0.03 \text{ m}^2/\text{g}$), and kaolinite ($10.97 \pm 0.33 \text{ m}^2/\text{g}$).

Mixtures were prepared by combining the metallic or clay constituent with Ottawa sand. Nine iron mixtures were prepared (0.50%, 1%, 5%, 10%, 20%, 35%, 50%, 75%, and 100% iron by weight). Nine magnetite mixtures were also prepared (0%, 5%, 15%, 30%, 50%, 70%, 85%, 95%, and 100% magnetite by weight). Saturation of samples with kaolinite concentration greater than 35% was hindered by the low permeabilities resulting from the dispersion of clay particles throughout the sand matrix. Therefore, twelve mixtures were prepared from 0% to 32% kaolinite by weight (0%, 1.3%, 1.6%, 2%, 2.5%, 3.2%, 4%, 5%, 6.3%, 8%, 10%, and 32%). Mixtures were dry-packed into cylindrical sample holders (16 cm $\times$ 3 cm ID). Samples were prepared as nearly identically as possible with respect to packing and composition. Mixtures were purged with CO_2 and evacuated (at -1.5 psi) prior to saturation with a 0.01 mol NaNO_3 solution.

Electrical measurements were made with a dynamic signal analyzer (DSA). The conductivity magnitude ($|\sigma|$) and phase angle (ϕ) were measured relative to a resistor at 40 frequency intervals between 10^{-1} and 10^3 Hz (10^{-1} and 10^4 Hz for the magnetite mixtures). Current was injected via stainless

steel electrodes placed at either end of the column. Potential measurements were recorded using silver-silver chloride electrodes. The real (σ') and imaginary (σ'') conductivity were calculated from the conductivity magnitude ($|\sigma|$) and phase angle (ϕ) measurements. Instrumentation errors were reduced by calibrations on water samples of known complex conductivity ($\sigma^*(\phi)$). Between 0.5 and 100 Hz, errors are generally less than $\pm 0.5 \text{ mrad}$ for ϕ and $\pm 0.5\%$ for $|\sigma|$ respectively. Capacitive coupling at the potential electrode connection increases the errors in ϕ above 100 Hz. For details of instrumentation and performance, see Slater and Lesmes (2002).

At the end of the experiment, saturation and porosity (ϕ) were determined from weight loss between saturated and dried samples. Saturation exceeded 0.96 for all samples. For each mixture S_p (μm^{-1}) was calculated using

$$S_p = [(S_{s_x} \times \rho_x \times W_x) + (S_{s_{OS}} \times \rho_{OS} [1 - W_x])] \times \frac{1 - \phi}{\phi} \quad (1)$$

where ρ is density (g/cm^3), S_s is m^2/g , and W is fractional weight; subscript OS refers to the Ottawa sand and subscript x refers to the mineral (iron, magnetite, or kaolinite) in the mixture.

A Cole-Cole (Cole and Cole, 1941) relaxation was used to fit data for the iron and magnetite sample. This was formulated in terms of complex conductivity (Jones, 2002) as

$$\sigma^*(\omega) = \sigma_0 \left[1 + m \left(\frac{(i\omega\tau)^c}{1 + (i\omega\tau)^c(1 - m)} \right) \right] \quad (2)$$

where σ_0 is the conductivity at DC frequency, τ is the mean relaxation time, c is a shape exponent that typically ranges from 0.1 to 0.6, and m is the chargeability (given by $m = 1 - \sigma_0/\sigma_\infty$, where σ_∞ is the conductivity at high frequency). The parameter $m_n = m \times \sigma_0$, represents a global measure of polarization accounting for the frequency dependence of σ^* (Lesmes and Frye, 2001). We used an algorithm employing a least-squares approach with Marquardt regularization (Kemna, 2000).

RESULTS

Figure 1 shows representative spectral data sets obtained for three mixtures: 10% iron, 15% magnetite, and 5% kaolinite. The iron and magnetite mixtures can be fit with a Cole-Cole model, as illustrated for the iron. Although the kaolinite data set could also be fit to a Cole-Cole model, it can be approximated by a power-law dependence of saturation and porosity (ϕ) on frequency. Since the power-law exponent does not vary greatly between the 12 kaolinite data sets (average = 0.24, standard deviation = 0.04), a single-frequency measurement of σ'' [e.g., at 1 Hz ($\sigma''_{1 \text{ Hz}}$)] represents the polarization magnitude of these mixtures. The linear correlation coefficient (R^2) is shown for the fit of the kaolinite data set.

The polarization magnitude as a function of S_p for all three mixtures is shown in Figure 2. The iron and magnetite mixtures are plotted in terms of m_n derived from Cole-Cole modeling, whereas the kaolinite mixtures are plotted in terms of $\sigma''_{1 \text{ Hz}}$. Power-law fits to the data sets show that the relationship between polarization magnitude and S_p is almost linear. A single power-law relationship is able to fit the magnetite and iron mixtures (exponent of 1.03). The exponent is 0.96 for the kaolinite mixtures.

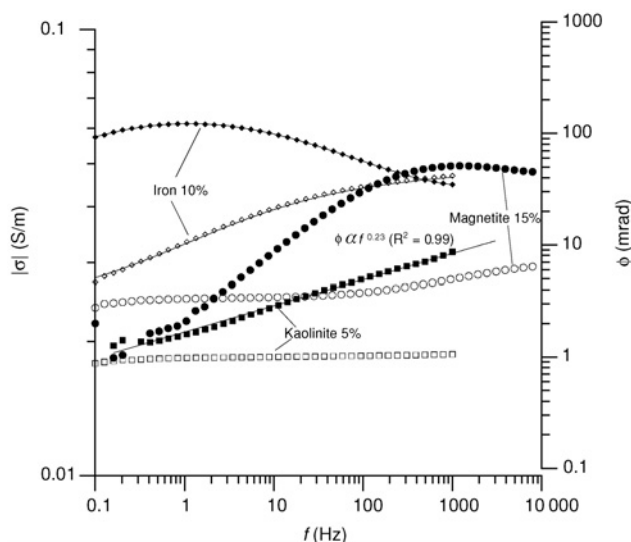


Figure 1. Spectral data sets obtained for three mixtures: 10% iron, 15% magnetite, 5% kaolinite. Solid symbols are the phase (ϕ) and open symbols are the conductivity magnitude ($|\sigma|$). The fit of the iron data set to a Cole-Cole model ($\sigma_0 = 0.0271 \text{ S/m}$; $m = 0.51$; $\tau = 0.33 \text{ s}$; $c = 0.424$) is shown. The fit of the kaolinite data set to a power-law expression is also shown. Note that the $|\sigma|$ axis covers only one order of magnitude, whereas the ϕ axis covers four orders of magnitude [frequency (f) dependence of $|\sigma|$ is thus greatly exaggerated]. Linear correlation coefficient (R^2) is shown for the kaolinite data set.

The dependence of the global measure of polarization (m_n) and imaginary conductivity ($\sigma''_{1\text{ Hz}}$) on pore volume (S_p) for the iron and magnetite mixtures is contrasted in Figure 3. Here m_n data for both mixtures are shown as a single data set since they follow the same near linear relation with S_p (Figure 2). Two differences arise in the $\sigma''_{1\text{ Hz}}$ dependence on S_p between the mixtures: (1) The magnitude of $\sigma''_{1\text{ Hz}}$ is greater for the iron mixtures, and (2) the relationship is approximately linear for the magnetite mixtures (exponent of 1.07) but not for the iron mixtures (exponent of 1.6).

DISCUSSION

Measures of polarization magnitude in these metal-sand and clay-sand mixtures ($R^2 > 0.93$ in every case) show a strong power-law dependence on S_p . The relationship is near linear except for $\sigma''_{1\text{ Hz}}$ versus S_p in the sand-iron mixtures where the power-law exponent is 1.6. The frequency of the dispersion peak in this data set is close to 1 Hz, and the position of this peak in frequency (hence Cole-Cole τ) varies with S_p (Slater et al., 2005, Figure 2). This effect increases the dependence of $\sigma''_{1\text{ Hz}}$ on S_p , increasing the power-law exponent. The dispersion peak in the magnetite data set occurs above 100 Hz; hence, shifts in this peak exert minimal control on $\sigma''_{1\text{ Hz}}$. In the magnetite mixture, the higher-frequency polarization peak also causes $\sigma''_{1\text{ Hz}}$ to be an order of magnitude less than the iron mixture at equivalent S_p (Figure 3). Using a global measure of the polarization (accounting for the frequency dependence) computed from Cole-Cole modeling, a

single, near-linear dependence on S_p is found for both data sets.

These results conflict with traditional concepts of IP in soils. Textbooks typically refer to electrode polarization associated with metals blocking electrolyte-filled pore passages (e.g., Reynolds, 1997, Figure 9.5). In our samples, iron filings dispersed within Ottawa sand do not block pore spaces, and the mixture porosity increases with the percentage iron (from 0.30 for 0% iron to 0.38 for 100% iron). Our results show that the polarization simply scales linearly in relation to the amount of the interface in contact with pore fluid (as quantified by S_p) and does not imply pore blockages.

We made IP measurements on iron particles that were suspended in a solution by attaching them to tape. Although $\phi < 5$ mrad, Cole-Cole dispersions were observed at IP frequencies (data not shown for brevity). Similar arguments pertain to the kaolinite mixtures. The traditional concept of membrane polarization invokes formation of cation-selective membranes impeding electrolytic conduction through pores. A common belief is that membrane polarization is strongest at relatively low concentrations of clays that are well dispersed throughout a soil. We find the polarization to scale near linearly with S_p up to 32% kaolinite concentration.

We observed a similar dependence of polarization magnitude on S_p for both metal and clay mixtures even though the polarization mechanisms at the interface are somewhat different. Note the very similar $\sigma''_{1\text{ Hz}}$ versus S_p relationships for the clay (Figure 3) and the magnetite (Figure 4) mixtures. In the clay mixtures, ionic diffusion occurs predominantly along the mineral surface. The metallic minerals in the iron and magnetite mixtures are electronic conductors, so ions migrate to/from the conductive interface (Wong, 1979). Furthermore, redox-active ions contribute an electrochemical polarization

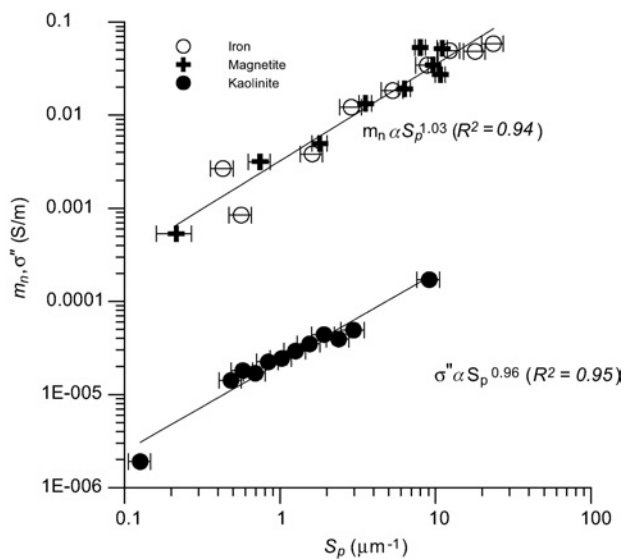


Figure 2. Polarization magnitude as a function of S_p for the sand-iron, sand-magnetite, and sand-clay mixtures. Polarization magnitude for the sand-iron and sand-clay mixtures is defined as m_n obtained from Cole-Cole modeling. A local measure of the polarization magnitude ($\sigma''_{1\text{ Hz}}$) is plotted for the sand-kaolinite mixtures. Power-law relations for 1) the combined sand-iron and sand-magnetite data set and 2) the sand-kaolinite data set are shown. Error bars are plotted when larger than symbol size. Linear correlation coefficients (R^2) are shown.

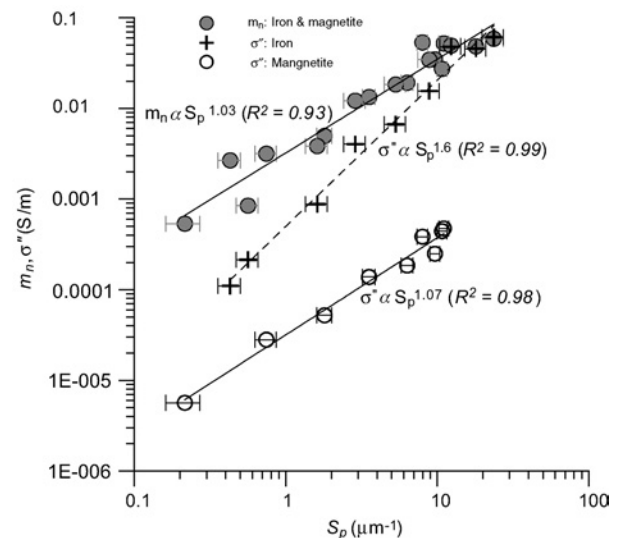


Figure 3. Local ($\sigma''_{1\text{ Hz}}$) and global (m_n) measures of the polarization magnitude for iron-sand and magnetite-sand mixtures as a function of S_p . The mixtures are shown as a single m_n data set, since they follow the same trend. Error bars are plotted when larger than symbol size. Linear correlation coefficients (R^2) are shown.

that facilitates conversion between electrolytic and electronic conduction at the interface. These differences are revealed in the high m_n associated with the metal mixtures (Figures 1 and 2) but not in the dependence on S_p .

Natural soils and rocks are more complex than the mixtures studied here because pore-scale properties other than S_p , i.e., mineral-fluid interface geometry, potentially affect the IP of natural soils. Previous studies adopted an interfacial-formation factor in empirical equations used to describe the IP response of soils and rocks (Vinegar and Waxman, 1984) (Börner et al., 1996). Scott and Barker (2005) show m_n as a function of S_p for sixteen sandstones (wide range of mineralogy, porosity, and S_s) from five formations within the United Kingdom. Their results are reproduced in Figure 4: The power-law relationship ($R^2 = 0.66$) between m_n and S_p is nonlinear (exponent of 0.5). Scott and Barker attribute the weak relationship between m_n and S_p to differences in the distribution of clay minerals throughout the sand matrix between samples. A second data set for natural soils found in Slater and Glaser (2003) is reproduced in Figure 4. These sands and silts from the Kansas City River floodplain exhibit limited variation in S_p (less than an order of magnitude) and a nonlinear power-law relationship ($R^2 = 0.6$) between σ'' and S_p (exponent of 1.2).

Although petrophysical properties other than S_p undoubtedly affect the IP of natural soils and rocks, we recognize an alternative explanation for the discrepancy between our results and others reported for natural soils. For the two constituents of each mixture, we measured S_s and then calculated the effective S_s of the sample using equation 1. Assuming the constituents were well mixed, S_s from the BET measurement should be representative of S_s at the scale of the IP measurement. In contrast, BET measurements on soils are typically

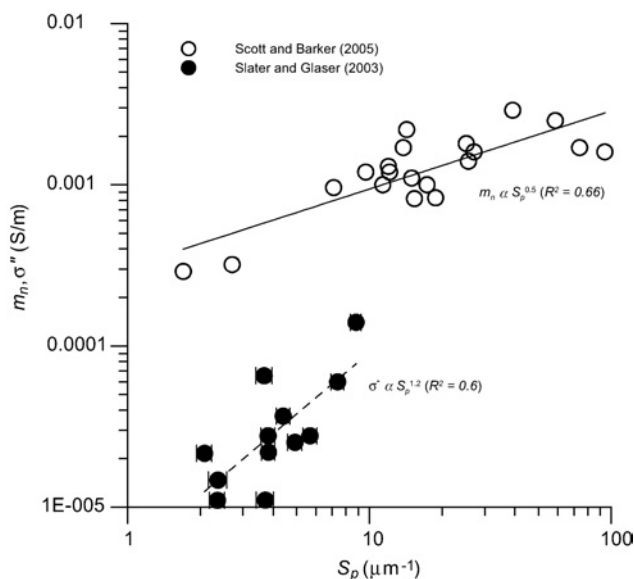


Figure 4. Power-law relationships between polarization magnitude and S_p observed in natural soils/rocks. In the sandstone data from Scott and Barker (2005), m_n is used to define the polarization magnitude. In the sandy alluvial sediments from Slater and Glaser (2003), $\sigma''_{1\text{ Hz}}$ is used to define the polarization magnitude. Linear correlation coefficients (R^2) are shown.

made on a small subsample of the soil used for electrical measurements. We must then question whether the BET-derived S_s adequately reflects the S_s at the scale of the IP measurement. For example, a small amount of clay or metal within a soil can profoundly influence the polarization magnitude. If this clay or metal is not present in identical concentrations in the subsample extracted for BET measurement, then the correct dependence of polarization magnitude on BET-derived S_p naturally cannot be measured.

In our study, measurements were made at the same electrolyte conductivity. The effect of electrolyte chemistry on IP remains uncertain; however, it appears to exert a relatively minor control in nonmetallic soils over a conductivity range typical of groundwater (Lesmes and Frye, 2001). The IP dependence on electrolyte chemistry is greater in metallic soils (Slater et al., 2005).

Although the IP response of soils remains an uncertain function of the physicochemical properties of soil and electrolyte, it is clear that S_p is a critical petrophysical property of the soil determining IP. The relationship between polarization magnitude and S_p is comparable for metallic and non-metallic soils. Similar near-linear relationships between measures of the interfacial polarization and S_p are observed at kHz to MHz frequencies (Knight and Nur, 1987), suggesting that common mechanisms may control electrical properties across a wide frequency range. Current efforts to develop theoretical models to describe IP in soils clearly should incorporate this dependence on S_p .

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